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Department of Computer Application (BCA)
BCAC501 — PHP with MySQL | Semester V

PROJECT DETAILS REPORT

1. Project Title

Lab Equipment Borrowing System

2. Problem Statement

Students and faculty regularly need to borrow shared lab equipment — Arduino kits, projectors, HDMI cables, extension boards, and lab manuals — but there is currently no organized way to track who has what, when it is due back, or whether an item is even available. This leads to lost items, double-booking, and no accountability when equipment is returned late or damaged. This system lets students and faculty browse available equipment and request to borrow it for a specific time window, while the lab in-charge (Admin) approves requests, tracks issued items, and manages returns — giving both sides a clear, accountable record.

3. User Roles

The system has two roles, matching the minimum requirement set in the project notice:

- Student / Faculty (Borrower) — Registers/logs in, browses the equipment catalog, submits a borrow request for a date range, views the status of their own requests, and edits their own profile/contact details.
- Admin (Lab In-Charge) — Logs in separately, manages the equipment catalog (add/edit/remove items and available quantity), views ALL borrow requests from every user, approves or rejects requests, marks items as Issued when handed over and Returned when given back, and can see overdue items at a glance.

4. Module List — Mapped to This Project

Every mandatory module from the project notice, mapped specifically to this project's subject matter:

#	Mandatory Module	How It Applies to This Project
1	Authentication	Student/Faculty Register & Login, and a separate Admin Login. Passwords stored using password_hash().
2	Role-Based Access	Two roles — Borrower and Admin — each with a different dashboard and different permissions after login.
3	Dashboard	Borrower dashboard shows their active/pending requests and quick 'Browse Equipment' shortcut. Admin dashboard shows pending-approval count, currently-issued count, and overdue count.
4	Core Entity CRUD	The core entity is Equipment. Admin can Add / View / Edit / Delete equipment items (name, category, total quantity, available quantity).
5	Listing Page	Equipment Catalog page — all items in a table/grid with Search (by name) and Filter (by category / availability).
6	Detail View	Clicking a Borrow Request opens a detail page showing full history: who requested, what item, requested dates, approval date, issue date, return date.
7	Status Workflow	Each borrow request moves through 4 states: Pending -> Approved -> Issued -> Returned (or Rejected instead of Approved). Only Admin can change status.

8	Admin Panel	A separate Admin area listing ALL requests (not just one user's), with buttons to Approve/Reject, mark Issued, mark Returned, and a highlighted Overdue view.
9	Form Validation	Borrow Request form validates that the return date is after the borrow date, that the requested quantity does not exceed availability, and that no field is left empty — checked in PHP, not just HTML5.
10	Flash Messages	Every action shows a message: request submitted, request approved/rejected, item marked returned, login failure, etc.
11	Profile Management	Borrower can view/edit their own name and contact number, needed so Admin can reach them about overdue items.

Optional modules included, since they genuinely apply to this project:

- Booking/Selection Flow — requesting an item for a specific date range is the core transaction of this system.
- Basic Reports/Counts — Admin dashboard shows counts of pending, issued, and overdue items.
- File Handling (covers Experiment 8) — three concrete features: (1) Borrower downloads a text/PDF receipt for an approved or issued request, generated server-side and served as a file download; (2) Admin exports the full requests list (or equipment list) to a CSV/Excel file for record-keeping; (3) Admin bulk-uploads new equipment by uploading a CSV file, which PHP reads row-by-row and inserts into the equipment table instead of adding items one at a time.

5. Database Tables (Draft)

The following is the current draft structure. Exact column types/sizes may be refined once development begins.

Table: users

Column	Type / Notes
user_id	INT, Primary Key, Auto Increment
name	VARCHAR(100)
email	VARCHAR(100), Unique
phone	VARCHAR(15)
password	VARCHAR(255) — stored using password_hash()
role	ENUM('borrower', 'admin')
created_at	DATETIME

Table: equipment

Column	Type / Notes
equipment_id	INT, Primary Key, Auto Increment
name	VARCHAR(100)
category	VARCHAR(50) — e.g. Electronics, AV, Accessories
total_quantity	INT
available_quantity	INT — decremented on approval, incremented on return
added_by	INT, Foreign Key -> users.user_id (Admin)

Table: borrow_requests

Column	Type / Notes
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request_id	INT, Primary Key, Auto Increment
user_id	INT, Foreign Key -> users.user_id
equipment_id	INT, Foreign Key -> equipment.equipment_id
quantity	INT
requested_from	DATE
requested_to	DATE
status	ENUM('pending','approved','rejected','issued','returned')
requested_at	DATETIME
approved_at	DATETIME, Nullable
issued_at	DATETIME, Nullable
returned_at	DATETIME, Nullable

6. Page List

#	Page / Screen	Accessible By	Purpose
1	Register	Public	New borrower account creation
2	Login (Borrower)	Public	Borrower authentication
3	Login (Admin)	Public	Separate Admin authentication
4	Borrower Dashboard	Borrower	Landing page — summary of own requests, quick links
5	Equipment Catalog (Listing)	Borrower	Browse/search/filter all available equipment
6	Equipment Detail	Borrower	View single item's details, submit a borrow request
7	My Requests (Listing)	Borrower	List of the borrower's own requests with current status
8	Request Detail View	Borrower / Admin	Full history of a single borrow request
9	Profile	Borrower	View/edit own name, contact info, change password
10	Admin Dashboard	Admin	Pending/Issued/Overdue counts, quick links
11	Manage Equipment (Listing + Add/Edit/Delete)	Admin	Core entity CRUD for equipment items
12	All Requests (Admin Listing)	Admin	Every borrow request from every user, with Approve/Reject/Issue/Return actions
13	Overdue Items	Admin	Filtered view of requests past their return date and not yet returned
14	Download Receipt	Borrower	Generates and downloads a text/PDF receipt for an approved or issued request
15	Export Requests (CSV/Excel)	Admin	Exports all borrow requests to a CSV file for record-keeping
16	Bulk Upload Equipment	Admin	Upload a CSV file of multiple equipment items; PHP reads and inserts each row

7. File Structure

lab-equipment-borrowing/

